

Literature Status Note: The Mixed-Norm Matrix Greedy-Basis Problem

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Status. `literature_already_answered`. The question in Dilworth–Freeman–Odell–Schlumprecht [1] is answered negatively by Schechtman [2].

Source question. In the introduction of [1], immediately after Theorem 2a, the authors write:

Assume $1 < p \neq q < \infty$. Does $\ell_q(\ell_p) = (\oplus \ell_p)_{\ell_q}$ have a greedy basis?

This is the extracted target from arXiv:0910.3867.

Supporting answer. Schechtman [2] explicitly identifies the same question. In the introduction he writes that Dilworth, Freeman, Odell and Schlumprecht raised the question whether

$$\left(\bigoplus_{n=1}^{\infty} \ell_p \right)_{\ell_q}, \quad 1 < p \neq q < \infty,$$

has greedy bases, and then states that these spaces do not have greedy bases. Theorem 1 in [2] proves the stronger assertion that each normalized unconditional basis of

$$\left(\bigoplus_{n=1}^{\infty} \ell_p \right)_{\ell_q}, \quad 1 \leq p \neq q < \infty,$$

has one subsequence equivalent to the unit vector basis of ℓ_p and another equivalent to the unit vector basis of ℓ_q . Consequently none of these spaces has a greedy basis.

Why this answers the source problem. The source problem is exactly the reflexive subrange $1 < p \neq q < \infty$ of Schechtman's Theorem 1. The answer is therefore negative: $\ell_q(\ell_p)$ does not have a greedy basis when $1 < p \neq q < \infty$. The supporting author explicitly cites the source paper and states that this solves the problem, so this packet is a literature-status record rather than a new proof packet.

Scope limitations. No part of the stated source problem remains open in this packet. Schechtman's additional endpoint conclusions for mixed c_0 variants are not part of the source question recorded here.

Search evidence. The local parsed source for arXiv:0910.3867 contains the problem immediately after Theorem 2a. The local parsed source for arXiv:1310.2371 contains the explicit DFOS citation, the phrase that the paper solves the raised question, and Theorem 1. Web search for the exact phrases "`ell_q(ell_p)`" "`greedy basis`" and "`Greedy bases for Besov spaces`" "`ell_q(ell_p)`" returned arXiv:1310.2371 as the direct answer.

References

- [1] S. J. Dilworth, D. Freeman, E. Odell, and Th. Schlumprecht, *Greedy bases for Besov spaces*, arXiv:0910.3867.
- [2] Gideon Schechtman, *No greedy bases for matrix spaces with mixed ℓ_p and ℓ_q norms*, arXiv:1310.2371.